

# Acoustic Violin Build Packet

## Instrument-Maker Print Packet

Build packet folder: /tmp/acoustic-violin-codex-gina-r2-build-packet

Generated: 2026-05-09

This packet is the printable companion to the build folder. Take it shopping or into the shop. Tear sheets at page breaks.

### File Map

| File                | Purpose                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------|
| design.md           | Project intent, catalog metadata, assumptions, and validation plan.                       |
| bom.csv             | Starter bill of materials with part categories, quantities, drawing refs, and notes.      |
| sourcing.csv        | Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks. |
| cut-list.csv        | Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.     |
| drawing-brief.md    | Manufacturing drawing and technical product sketch brief.                                 |
| assembly-manual.md  | Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.  |
| validation.csv      | Target/measured values, tolerance, environment, result, and tuning/build action log.      |
| supplier-rfq.md     | Supplier email/request-for-quote starter.                                                 |
| visual-bom-brief.md | Art direction for an image-forward visual BOM.                                            |
| wolfram-starter.wl  | Wolfram starter for physics, optimization, visualization, and validation.                 |
| README.md           | Project artifact.                                                                         |
| photo-shotlist.md   | Project artifact.                                                                         |
| risks.md            | Project artifact.                                                                         |

# design.md

*Project intent, catalog metadata, assumptions, and validation plan.*

## Acoustic Violin Design

### Intent

Create an honest L2 acoustic violin packet that captures the standard setup geometry and the open engineering gaps before any claim of build readiness.

### Readiness

L2 scaffold. Use for design review, drawing planning, and sourcing review. Do not treat this as a final luthiery method or validated arching plan.

### Governing Model

Open strings use Mersenne-Taylor. Body response is not reduced to a simple closed-form model in this packet; arching, graduation, bridge, soundpost, and air/body modes must be validated empirically.

$$f = (1 / (2 L)) * \text{sqrt}(T / \mu)$$

### Baseline Assumptions

|                       |                              |                                  |
|-----------------------|------------------------------|----------------------------------|
| Parameter             | Baseline                     | Notes                            |
| ---                   | ---:                         | ---                              |
| Scale length          | 328 mm / 12.91 in            | Standard violin vibrating length |
| Tuning                | G3-D4-A4-E5                  | A4 = 440 Hz reference            |
| String set            | Standard 4-string violin set | Use published tensions           |
| Body length           | about 14 in / 356 mm         | Exact outline TBD from drawing   |
| Top/back              | Spruce top, maple back       | Graduation not finalized         |
| Neck angle/projection | TBD from setup drawing       | Critical to bridge height/action |
| Target total tension  | about 45-55 lbf              | Verify with selected string set  |

### Tension And Setup Notes

- Fit bridge and soundpost under low tension, then bring strings up gradually.

- Check neck projection before final bridge height.
- Record action, bridge foot fit, soundpost location, and open-string pitch stability.

## Known Gaps

- No arching/graduation map.
- No neck-set drawing.
- No soundpost/bridge fitting evidence.
- No live sourceability check.
- No acoustic measurements.

# bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

| item                      | qty | unit | material_or_spec                                    | estimated_cost_usd | notes                                 |
|---------------------------|-----|------|-----------------------------------------------------|--------------------|---------------------------------------|
| Top plate                 | 1   | ea   | spruce violin top blank or prototype plate          | 90                 | Graduation TBD                        |
| Back plate                | 1   | ea   | maple back blank or prototype plate                 | 110                | One-piece or two-piece                |
| Rib/lining/block stock    | 1   | set  | maple ribs, spruce blocks/linings                   | 45                 | Bending method TBD                    |
| Neck blank                | 1   | ea   | maple violin neck blank                             | 55                 | Scroll detail optional for prototype  |
| Fingerboard               | 1   | ea   | ebony or composite violin fingerboard               | 35                 | Final scoop/setup TBD                 |
| Bridge blank              | 1   | ea   | violin bridge blank                                 | 12                 | Must be fitted to top                 |
| Soundpost                 | 1   | ea   | spruce soundpost stock                              | 5                  | Fit and position are validation items |
| Tailpiece/chinrest/endpin | 1   | set  | standard violin fittings                            | 45                 | Compatibility TBD                     |
| Pegs or tuners            | 4   | ea   | violin pegs or geared pegs                          | 35                 | Pegbox geometry must match            |
| String set                | 1   | set  | standard violin G-D-A-E set                         | 45                 | With published tension                |
| Finish/adhesive           | 1   | lot  | hide glue or prototype adhesive; varnish system TBD | 45                 | Traditional methods TBD               |

# sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

| item            | required_spec                          | search_terms        | supplier_candidates                | dates_checked      | unit_price_usd | lead_time | substitution_risk | notes                           |
|-----------------|----------------------------------------|---------------------|------------------------------------|--------------------|----------------|-----------|-------------------|---------------------------------|
| Violin strings  | 4/4 violin set with published tensions | violin strings      | string manufacturers               | EB Music suppliers | TBD            | TBD       | Low               | Use actual tension values       |
| Bridge blank    | 4/4 violin bridge blank                | violin bridge blank | luthier suppliers                  | TBD                | TBD            | TBD       | Low               | Quality affects setup           |
| Soundpost stock | straight spruce soundpost              | violin soundpost    | luthier suppliers                  | TBD                | TBD            | TBD       | Low               | Fit is more important than cost |
| Tonewood plates | spruce top and maple back              | violin plates       | luthier suppliers                  | TBD                | TBD            | TBD       | Medium            | Prototype may use lower grade   |
| Fittings        | tailpiece, chinrest, endpin            | violin fittings     | luthier suppliers; music suppliers | TBD                | TBD            | TBD       | Medium            | Peg/tuner choice affects pegbox |

# cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

| part           | qty | rough_dimensions       | final_dimensions    | material        | grain_or_orientation    | operation             | notes                      |                   |
|----------------|-----|------------------------|---------------------|-----------------|-------------------------|-----------------------|----------------------------|-------------------|
| Top plate      | 1   | 0.75 x 8.5 x 15.0      | arched/graduated    | spice           | quarter-sawn center     | glinecarve/CNC, grade | Arching map required       |                   |
| Back plate     | 1   | 0.75 x 8.5 x 15.0      | arched/graduated    | TBD             | quarter-sawn center     | glinecarve/CNC, grade | Graduation TBD             |                   |
| Ribs           | 6   | 0.060 x 1.5 x variable | profile TBD         | maple           | long grain lengthwise   | band or laminate      | Mold required              |                   |
| Blocks/linings | 1   | set                    | TBD                 | TBD             | spruce/willow/linings   | stock                 | shape and glue             | Corner block plan |
| Neck           | 1   | 2.0 x 2.5 x 11.0       | violin neck profile | TBD             | quarter grain preferred | saw, carve, fit       | Projection critical        |                   |
| Fingerboard    | 1   | 0.35 x 1.75 x 11.0     | scoop/profile TBD   | ebony/composite | long grain              | plane and fit         | Final setup after neck set |                   |

TBD

# drawing-brief.md

*Manufacturing drawing and technical product sketch brief.*

## Acoustic Violin Drawing Brief

Required first drawings:

- Full outline with centerline, f-hole placeholders, bridge line, and stop.
- Side setup drawing with nut, fingerboard projection, bridge, and tailpiece.
- Plate arching/graduation placeholder map with TBDs visible.
- Rib garland and block layout.
- Neck/pegbox/fingerboard interface detail.

# assembly-manual.md

*Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.*

## Acoustic Violin Assembly Manual

1. Choose the outline, arching, and graduation references before cutting.
2. Prepare plates, ribs, blocks, linings, neck, and fingerboard.
3. Build the rib garland on a mold or documented fixture.
4. Carve or CNC rough plates, then hand-finish critical arching and graduation.
5. Close the body with reversible or documented adhesive.
6. Fit neck angle/projection before final fingerboard work.
7. Fit pegs/tuners, nut, bridge, soundpost, tailpiece, and strings.
8. Bring strings to pitch gradually and log setup measurements.

Safety: string and bridge forces are lower than a harp but still require eye protection and careful first pitch-up.



# validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

| check_id | area            | target                              | method                         | tolerance              | measured | result | action                    |
|----------|-----------------|-------------------------------------|--------------------------------|------------------------|----------|--------|---------------------------|
| VAL-001  | Scale length    | 328 mm / 12.91 in vibration length  | measure length to bridge       | +/- 1 mm               | TBD      | TBD    | Reset bridge/nut position |
| VAL-002  | String tension  | within selected set published range | visual distance chart          | within published range | TBD      | TBD    | Change string set         |
| VAL-003  | Neck projection | setup-compatible bridge height      | measure bridge to bridge plate | target TBD             | TBD      | TBD    | Reset neck or bridge plan |
| VAL-004  | Action          | playable G/E string height          | measure at fingerboard         | target TBD             | TBD      | TBD    | Adjust bridge/fingerboard |
| VAL-005  | Bridge fit      | full foot contact on top            | visual light-gap check         | no visible gap         | TBD      | TBD    | Refit bridge              |
| VAL-006  | Soundpost fit   | stable post without crimp           | inspection/backbone test       | no slipping/buzzing    | TBD      | TBD    | Refit/reposition          |

## supplier-rfq.md

*Supplier email/request-for-quote starter.*

### Supplier RFQ Draft

Hello,

Please quote materials for a 4/4 acoustic violin prototype: spruce top blank, maple back/neck/rib stock, bridge blank, soundpost stock, standard fittings, and a 4-string set with published tension data.

Please include dimensions, grade, unit price, minimum order quantity, lead time, and substitution options.

## visual-bom-brief.md

*Art direction for an image-forward visual BOM.*

### Visual BOM Brief

Show the violin as assembled and as an exploded plate/rib/neck/fittings view.

Call out top, back, ribs, blocks, linings, neck, fingerboard, bridge, soundpost, tailpiece, chinrest, pegs/tuners, strings, adhesive, and finish.

# wolfram-starter.wl

*Wolfram starter for physics, optimization, visualization, and validation.*

(\* Acoustic violin open-string starter. Values are first-pass assumptions. \*)

```
ClearAll["Global`*"];
```

```
a4 = 440;
```

```
freqFromMidi[m_] := a4*2^((m - 69)/12);
```

```
scaleLengthIn = 12.91;
```

```
tuning = <|"G3" -> 55, "D4" -> 62, "A4" -> 69, "E5" -> 76|>;
```

```
Dataset[
```

```
KeyValueMap[
```

```
<|"string" -> #1, "frequencyHz" -> N[freqFromMidi[#2]],
```

```
"scaleLengthIn" -> scaleLengthIn|> &,
```

```
tuning
```

```
]
```

```
]
```

# README.md

*Project artifact.*

## Acoustic Violin

Engineering documentation and parametric design table for the acoustic violin - traditional construction reference and stave-built explorations.

Part of the [tonykoop/instrument-maker](https://github.com/tonykoop/instrument-maker) catalogue.

## Readiness

L2 scaffold. This repo now includes root-mode build packet documentation for a 4-string acoustic violin reference build. It is not L3 because arching, graduation, neck-set, soundpost, bridge setup, CAD/render evidence, and measured prototype validation are not complete.

## Packet Contents

- ``design.md`` - acoustic violin baseline and bowed-string assumptions.
- ``bom.csv``, ``sourcing.csv``, ``cut-list.csv``, ``validation.csv`` - packet tables.
- ``assembly-manual.md``, ``risks.md``, ``photo-shotlist.md`` - build and review aids.
- ``drawing-brief.md``, ``drawings/README.md``, ``cad/README.md``,  
``wolfram-starter.wl``, ``wolfram/README.md`` - starter technical artifacts.

## String-Scale Assumptions

Baseline: 4 strings, G3-D4-A4-E5, 328 mm / 12.91 in vibrating length, standard violin string set, and total tension verified against the manufacturer chart before first pitch-up.

## License

[CC BY 4.0](LICENSE) - see LICENSE for details.

## photo-shotlist.md

*Project artifact.*

### Photo Shotlist

- Plate blanks with grain orientation.
- Rib mold or garland setup.
- Arching/graduation checks.
- Neck projection straightedge before glue.
- Bridge foot fit and soundpost fit.
- First string-up with action and tuner readings.

## risks.md

*Project artifact.*

### Acoustic Violin Risks

- Arching and graduation are not validated and can dominate tone and structure.
- Neck projection errors can make the instrument unplayable.
- Soundpost and bridge fitting require care; poor fit can damage the top.
- Live sourcing and price checks are still missing.
- Remain at L2 until setup, CAD, and measured acoustic evidence exist.